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Understanding Mechanisms of PFAS Absorption, Distribution, and Elimination Using a Physiologically Based Toxicokinetic Model

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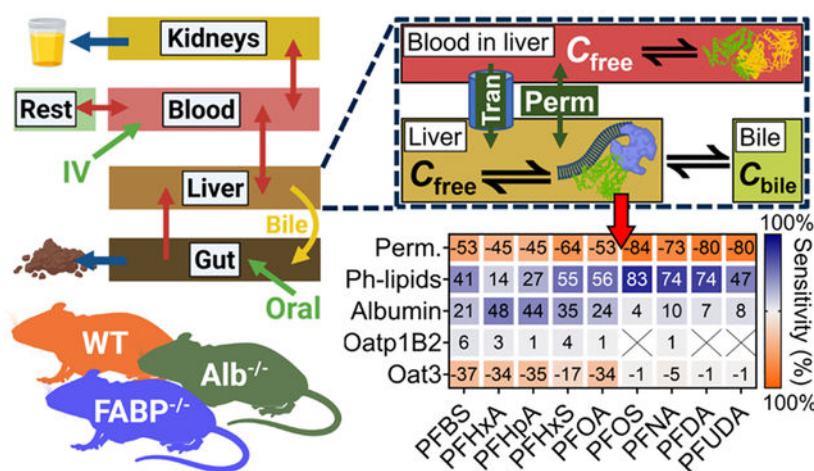
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Abstract

Exposures to some per- and polyfluoroalkyl substances (PFAS), including perfluoroalkyl acids (PFAA), have been associated with diverse adverse health effects. Physicochemical properties of PFAA are known to influence their toxicokinetics in mammals, but mechanistic models capable of identifying the key drivers of absorption, distribution, and elimination are limited. Here, we develop and evaluate a physiologically based toxicokinetic (PBTk) model parameterized to an *in vivo* mouse model using data from *in vitro* studies. We simulated tissue concentrations of 9 PFAA with perfluorinated carbon chains (η_{pfc}) ranging from 4 to 10 in wild-type mice, and for perfluorooctanesulfonic acid in knockout mice, following intravenous and oral exposures. The PBTk model quantifies blood flow, binding to tissue proteins and phospholipids, membrane permeability, hepatic and renal transporters, and fecal and urinary excretion. Model evaluation comparing experimental and simulated mice blood and tissue concentrations showed R^2 values of ≥ 0.65 and relative root-mean-square errors of $\leq 122\%$ for most PFAA. Model sensitivity analyses showed that permeability and phospholipid binding strongly influenced the elimination and distribution of long-chain PFAA ($\eta_{\text{pfc}} \geq 7$), while elimination for short-chain PFAA ($\eta_{\text{pfc}} \leq 6$) was more sensitive to renal transporters and albumin binding. This study illustrates how integrating *in vitro*-derived parameters into PBTk models enables mechanistic evaluation of PFAS toxicokinetics across diverse compounds and physiological conditions.

Graphical Abstract



Keywords

PFAS; tissue distribution; toxicokinetic modeling; elimination; mouse model

1. INTRODUCTION

Per- and polyfluoroalkyl substances (PFAS) are widely used in industrial and consumer applications for their surfactant and other properties.¹ Their ubiquitous presence in drinking water, food packaging, and consumer products² has resulted in detectable PFAS concentrations in 98–99% of U.S. individuals.^{3,4} PFAS exposure has been linked to

adverse effects on human health such as immunotoxicity,⁵ increased cancer risk,⁶ and metabolic disruption.⁷ However, factors governing the relationship between external PFAS exposures and internal concentrations at active sites where toxicity is elicited remain poorly understood.

Perfluoroalkyl acids (PFAA) are the most well-studied PFAS subclass. They include perfluorosulfonic acids (PFSA) such as perfluorooctanesulfonate (PFOS) and perfluorocarboxylic acids (PFCA) such as perfluorooctanoic acid (PFOA) and their precursors. Some PFAA with perfluorinated carbon chain lengths (η_{pfc}) of 7 or more have been shown to accumulate in humans, as evidenced by elimination half-lives of 3.5 years for PFOA ($\eta_{\text{pfc}} = 7$) and 4.8 years for PFOS ($\eta_{\text{pfc}} = 8$).⁸ In contrast, shorter chain PFAA ($\eta_{\text{pfc}} \leq 6$), such as perfluorobutanesulfonate (PFBS, $\eta_{\text{pfc}} = 4$), have elimination half-lives in humans ranging from days to weeks,⁹ consistent with findings from animal studies.^{10,11} In addition to physicochemical properties of PFAS, variability in measured human elimination half-lives reflect interindividual physiological differences driven by disease and/or genetic factors that are difficult to isolate. In contrast, time-dependent concentrations of PFAA in blood, organs, and excreta can be measured in physiologically comparable rodents following controlled doses.

Rodents are widely used in pharmaceutical and medical research to study pharmacokinetics.¹² Numerous studies have examined PFAA toxicokinetics in mice using single intravenous or oral doses to generate time dependent blood and tissue concentrations.^{13–17} Knockout mouse models, such as albumin- and fatty acid binding protein (FABP)-deficient C57BL mice, have been used to investigate specific toxicokinetic mechanisms affecting PFAA exposure. For example, albumin-deficient mice exhibited significantly elevated liver: blood concentration ratios,¹⁸ while FABP-deficient mice showed PFAA toxicokinetics similar to wild-type mice.¹⁹ Availability of physiological parameters for laboratory mouse strains, including organ volumes and blood flow rates,^{20,21} makes them a well-suited model species for mechanistic physiologically based toxicokinetic (PBTK) modeling.

Many important mechanisms affecting tissue PFAA distribution, elimination, and (re)absorption have been demonstrated *in vitro*. For example, PFAA with $\eta_{\text{pfc}} \geq 7$ show stronger binding to blood proteins such as albumin,²² and slower membrane permeability than their $\eta_{\text{pfc}} \leq 6$ counterparts.²³ Binding to membrane transporters such as organic anion-transporting polypeptides (Oatp) and organic anion transporters (Oat)^{23–26} is thought to drive PFAA uptake into liver and kidneys, respectively.²⁷ Multiyear elimination half-lives of $\eta_{\text{pfc}} \geq 7$ PFAA in humans have been attributed to enterohepatic recirculation and renal tubular reabsorption.^{28,29} PFAA with $\eta_{\text{pfc}} \geq 7$ are substrates for bile acid transporters such as apical sodium-dependent bile acid transporters (Asbt) and sodium/taurocholate cotransporting polypeptides (Ntcp).^{30,31} These transport mechanisms occur concurrently and interact dynamically under *in vivo* conditions and their overall importance for PFAA toxicokinetics requires further evaluation.

The main objective of this study was to better understand mechanisms affecting PFAS toxicokinetics in mammals. To do this, we developed a PBTK modeling framework that integrates *in vitro* data on permeability, tissue binding, and membrane transporters with physiological data for two mouse strains previously used in PFAS toxicokinetic studies: C57BL/6J (an inbred strain with minimal genetic variability) and CD-1 (an outbred strain with greater genetic diversity). We used the model to evaluate mechanistic drivers for variability in PFAA absorption, distribution, and elimination in wild-type and knockout mice for nine PFAA with $\eta_{\text{pfc}} = 4 - 10$. We conducted sensitivity analyses to identify the relative contributions of individual mechanisms to PFAA accumulation and elimination. We use these findings aim to gain insights into processes that are most important for PFAS toxicokinetics and highlight key knowledge and data gaps for extending findings to human populations and emerging PFAS.

2. MODEL THEORY AND PARAMETERIZATION

2.1. General Model Structure.

We developed a five-compartment PBTK model that includes: (1) central blood, (2) gut, (3) liver, (4) kidneys, and (5) the rest of the body (Figure 1). The “rest” compartment includes all other organs not specified. The model includes two exposure routes: oral PFAA ingestion through the gut lumen, and intravenous PFAA administration into the central blood compartment. Cells and interstitial fluid are represented as single organs, separated from perfusing blood by bidirectionally permeable membranes. The model accounts for: (a) blood flow to and from organs; (b) binding to proteins and phospholipids within organs; (c) passive membrane diffusion; (d) active transport via membrane transporters; (e) enterohepatic recirculation, (f) renal tubular reabsorption; and (g) excretion through feces and urine. Chemical parameters were derived from *in vitro* studies, while physiological parameters for CD-1 and C57BL mice were obtained from literature.

2.2. Description of the PBTK Model.

In mammals, physiological blood flow rates determine PFAA transport from central blood to organs. The model simulates temporal changes in blood and tissue concentrations ($dC/dt, \text{mol mL}^{-1} \text{ s}^{-1}$) based on chemical mass flows, representing PFAA absorption, distribution to the systemic circulation, distribution to organs, and elimination. PFAA bind to serum albumin, globulins, structural proteins, FABPs, and phospholipids in blood and organs, while binding to storage lipids is considered negligible.³² The concentration of freely dissolved PFAA ($C_{\text{free}}, \text{mol mL}^{-1}$) in blood is considered available for transport among organs.³³ PFAA bound to tissue components ($C_{\text{bound}}, \text{mol mL}^{-1}$) are not available for inter-organ transport and elimination. We simulated concentrations in whole blood rather than plasma to represent the full distribution of PFAS among plasma, cellular, and colloidal blood components.

We used desorption rate constants to account for PFAA dissociation from proteins and phospholipids.³⁴ Bidirectional chemical transport was assumed to occur via permeability. We incorporated transporter-mediated uptake and efflux of PFAA for the liver, gut, and

kidneys into the PBTK model to account for known transporter localization in mice. Hepatic uptake was modeled as unidirectional transport of free PFAA from blood into hepatocytes via Oatp1b2, Oatp2b1, and Ntcp. Enterohepatic recirculation was modeled by including biliary excretion into the intestinal lumen, followed by reabsorption via passive permeability and Oatp2b1-mediated uptake in enterocytes.²⁸ Renal elimination was modeled as a combination of glomerular filtration of free PFAA from the blood into the tubular lumen and Oat1, Oat2, and Oat3-mediated uptake of free PFAA from peritubular capillaries into proximal tubule cells,²⁹ followed by permeability across the apical membrane into the filtrate. The “rest” compartment was modeled as permeability-limited distribution without transporter involvement. Section S1 of the Supporting Information contains a complete description of model equations and parameter values.

2.2.1. Free and Bound Concentrations in Blood Perfusing Organ Tissues. The PBTK model simulates temporal changes in freely dissolved PFAA concentrations in blood perfusing each organ ($dC_{\text{free,blood}}/dt$, mol mL⁻¹ s⁻¹) by the net balance of mass flows

$$\frac{dC_{\text{free,blood}}}{dt} = (Q_{\text{free}}\Delta C_{\text{free,central-blood}} + B\Delta C_{\text{bound,blood-free,blood}} - P\Delta C_{\text{free,blood-tissue}} - TC_{\text{free,blood}})/(V_{\text{water,blood}}) \quad (1)$$

Where molar mass flow is a function of: (1) Free PFAA entering the blood perfusing the organ from the systemic circulation is described by $Q_{\text{free}}\Delta C_{\text{free,central-blood}}$, where Q_{free} (mL s⁻¹) is the perfusion-driven transport rate of free (unbound) PFAA and $\Delta C_{\text{free,central-blood}}$ (mol mL⁻¹) represents the concentration gradient of free PFAA between the central compartment and the organ blood. (2) Desorption from blood proteins and phospholipids is given by $B\Delta C_{\text{bound,blood-free,blood}}$, where B (mL s⁻¹) is the first-order desorption rate constant and $\Delta C_{\text{bound,blood-free,blood}}$ (mol mL⁻¹) is the concentration gradient between bound free PFAA in blood, defined as $C_{\text{bound,blood}} - C_{\text{free,blood}} \times K_{\text{blood/w}}$. (3) Permeability-driven transport into tissue is represented by $P\Delta C_{\text{free,blood-tissue}}$, where P ($P_{\text{app}} \times \text{surface area}$, mL s⁻¹) is the permeability-driven transport rate and $\Delta C_{\text{free,blood-tissue}}$ (mol mL⁻¹) is the concentration gradient of freely dissolved PFAA between blood and tissue. (4) Membrane transporter-mediated uptake is described by $TC_{\text{free,blood}}$, where T (mL s⁻¹) is the unidirectional uptake rate. $V_{\text{water,blood}}$ (mL) is the steady-state volume of water in the blood. For comparison with literature data, simulated blood concentrations were converted to plasma concentrations using a mass balance equation, as described in Section S1.

Temporal changes in concentrations of PFAA bound to blood proteins and phospholipids ($dC_{\text{bound,blood}}/dt$, mol mL⁻¹ s⁻¹) are described as

$$\frac{dC_{\text{bound,blood}}}{dt} = \frac{Q_{\text{bound}}\Delta C_{\text{bound,central-blood}} - B\Delta C_{\text{free-bound}}}{V_{\text{prot+lip,blood}}} \quad (2)$$

Transport of bound PFAA from the systemic circulation is given by $Q_{\text{bound}}\Delta C_{\text{bound,central-blood}}$, with $Q_{\text{bound}}(\text{mL s}^{-1})$ as the perfusion-driven transport rate of PFAA bound to blood proteins or phospholipids and $\Delta C_{\text{bound,central-blood}}(\text{mol mL}^{-1})$ as the concentration gradient of bound PFAA between the central compartment and the organ blood. $V_{\text{prot+lip,blood}}(\text{mL})$ is the total volume of blood proteins and phospholipids, including albumin, globulins, structural proteins, and phospholipids.

2.2.2. Free and Bound Concentrations in Organ Tissues. The model simulates temporal changes in freely dissolved concentrations of PFAA in organ tissues ($dC_{\text{free,tissue}}/dt, \text{mol mL}^{-1} \text{s}^{-1}$) as

$$\frac{dC_{\text{free,tissue}}}{dt} = (B\Delta C_{\text{bound,tissue-free,tissue}} + P\Delta C_{\text{free,blood-organ}} + T\Delta C_{\text{free,blood}} - k\Delta C_{\text{free,tissue-lumen}})/(V_{\text{water,tissue}}) \quad (3)$$

Desorption from tissue proteins and phospholipids is given by $B\Delta C_{\text{bound,tissue-free,tissue}}$, where $B(\text{mL s}^{-1})$ is the first-order desorption rate constant and $\Delta C_{\text{bound,tissue-free,tissue}}(\text{mol mL}^{-1})$ is the concentration gradient between bound and free PFAA in blood, defined as $C_{\text{bound,tissue}} - C_{\text{free,tissue}} \times K_{\text{tissue/w}}$. Elimination into the lumen is described by $k\Delta C_{\text{free,tissue-lumen}}$, where $k(\text{mL s}^{-1})$ is the clearance rate constant and $\Delta C_{\text{free,tissue-lumen}}(\text{mol mL}^{-1})$ is the concentration gradient between the tissue its respective lumen, which is the bile for liver, intestinal lumen for gut, and filtrate for kidneys. $V_{\text{water,tissue}}(\text{mL})$ is the steady-state volume of water in organ tissue.

Temporal concentration changes of PFAA bound to organ tissues ($dC_{\text{bound,tissue}}/dt, \text{mol mL}^{-1} \text{s}^{-1}$) are described as

$$\frac{dC_{\text{bound,tissue}}}{dt} = \frac{-B\Delta C_{\text{free-bound}}}{V_{\text{prot+lip,tissue}}} \quad (4)$$

$V_{\text{prot+lip,tissue}}$ is the volume of proteins and phospholipids in the tissue (L).

2.3. Physiological Parameters.

Physiological parameters for CD1 and C57BL/6J mice were sourced from the literature when available, extrapolated from rat studies when unavailable, or predicted based on

previous empirical data. Available physiological data included total body and organ weights for a 30 g mouse,^{35,36} steady-state blood volumes in organs,³⁵ blood and bile flow rates,^{35,37} capillary surface areas,^{38–40} excretion rates,^{41,42} and volume percentages of albumin, globulins, structural proteins, FABPs, and phospholipids in blood and organs.^{21,32,43–45} Section S2 provides detailed descriptions of data sources and a full list of physiological model input parameters. Physiological variability among mice was not incorporated into our simulation as it is thought to be small among laboratory mice of similar strain, age, and sex compared to the variability in chemical-specific input parameters.

2.4. Chemical Parameters.

Major categories of parameters in the PBTK model are: (1) PFAA apparent membrane permeabilities (P_{app}),²³ (2) partition coefficients for proteins and phospholipids,^{32,46–49} and (3) membrane transporter affinities.^{23–27,30,31,50} Model algorithms represent mechanistic processes that could be parameterized using reliable data derived in *in vitro* systems with available scaling factors. Only pathways with compound-specific data were included, and transporter rates were set to zero for compounds with no measurable activity or missing data. We used Monte Carlo simulations (10,000 iterations) to quantify the influence of measured variability/uncertainty in chemical parameters based on a detailed literature review of published *in vitro* studies (Tables S7–S10). We assumed log-normal distributions for partition coefficients and calculated μ and σ for each PFAS from the means and standard deviations of reported log K values (Table S8). For permeabilities and transporter affinities, we used normal distributions for each PFAS calculated from the grand mean and pooled variance calculated from replicate measurements within two *in vitro* studies,^{23,31} as summarized in Tables S9 and S10. Albumin desorption rates were used to account for desorption kinetics (1/s).⁵¹ Partition coefficients for phospholipids,^{47,48} serum albumin,^{46,49,52,53} and FABPs^{53–55} were obtained from multiple sources. Association and dissociation constants were transformed to partition coefficients following Endo et al.⁵⁶ *In vitro* permeabilities and transporter affinities were scaled to *in vivo* conditions using relative activity factors specific to the cell lines employed.⁵⁷ Section S3 of the Supporting Information provides additional information on chemical parameters, including their sources, variabilities, and methods for conversion and scaling.

2.5. Model Evaluation and Sensitivity Analysis.

Single and repeated intravenous and oral dosing studies for mice were used to evaluate model performance. We report the arithmetic mean of Monte Carlo simulations, and confidence intervals spanning the 2.5th and 97.5th percentiles of the ensemble for modeled time-dependent concentrations and elimination half-lives of each PFAA. Elimination half-lives were calculated by log-linear regression of the terminal phase blood concentration time curves for each simulation. We compared measured mean blood and tissue PFAA concentrations to the model expected values to calculate R^2 and relative root mean squared error (rRMSE). Specifically, we used data from three primary studies described below.

Modaresi et al.¹⁹ administered 0.1–5 mg/kg_{bw} PFOS to 5–6-month-old wild-type C57BL/6J mice and mice with local (liver, intestines) or global deletion of FABP. Blood was sampled

over 65 days, and liver, lung, kidney, intestine, brain, and muscle tissues were harvested at the end of the experiments. Kaye et al.¹⁸ administered daily oral doses of 0.5 or 10 mg/kg_{bw} PFOS to wild-type C57BL/6J mice and mice with global albumin deletion over a 7-day period. Plasma, liver, kidney, lung, and intestine samples were collected on day 7. Zhu et al.¹⁶ administered single oral or intravenous doses of 1 or 5 mg/kg_{bw} PFOA, PFOS, and perfluorohexanesulfonic acid (PFHxS) to male CD-1 mice, with plasma concentrations measured over 32 days.

We conducted a local sensitivity analysis by individually perturbing major parameters (permeability, partition coefficients, transporter rates) by 10% and quantifying the resulting changes in PFAA fractions absorbed, areas under the concentration–time curves (AUC) for blood and tissues, and elimination half-lives. Normalized sensitivity coefficients in percentages were calculated as the ratio of the relative change in toxicokinetic end points (absorption, distribution, elimination) to the relative change in model parameter, defined as

$$\text{normalized sensitivity}(\%) = \frac{(\Delta \text{outcome}/\text{outcome})}{(\Delta \text{parameter}/\text{parameter})} \times 100 \quad (5)$$

Parameters with the highest normalized sensitivity values were considered the most influential for PFAA absorption, distribution, and elimination. We implemented the PBTK model in *R* (v4.3.1) using the *deSolve* package. Monte Carlo sampling and global sensitivity analyses were performed using custom scripts and the *R* packages *mcmc*, *dplyr*, and *pracma*. The *R* code of the PBTK model is attached as a Supporting Information. All chemical and physiological input values were directly obtained from experimental studies or literature sources and were not calibrated to *in vivo* data.

3. RESULTS AND DISCUSSION

3.1. Model Evaluation.

We compared simulated plasma and blood concentrations of PFOS, PFOA, and PFHxS after single oral and intravenous (IV) doses to experimental data from C57BL/6J and CD-1 mice (Figure 2A–C). R^2 values ranged from 0.73 to 0.87 and rRMSE between 18% and 38% between modeled and observed values^{16,19} plasma and blood PFOS concentrations after single oral doses of 0.1–5 mg/kg_{bw}. Similarly, R^2 values ranged from 0.79–0.92 and rRMSE between 35%–41% between modeled and observed¹⁶ PFOA plasma concentrations after oral doses of 1 and 5 mg/kg_{bw}. We found lower R^2 values (0.045 and 0.01) and higher rRMSE values (65% and 70%) between modeled and observed¹⁶ PFHxS plasma concentrations at the same oral doses, primarily due to underestimation at early time points (Figure S3). For IV dosing of PFHxS, the model better matched experimental data with R^2 values of 0.71–0.90 and rRMSE ranging from 31% to 89%. However, plasma concentrations of PFOS and PFOA were underestimated following IV administration of 1 mg/kg_{bw}, with higher rRMSE values of 487% and 292% for PFOS and PFOA, respectively (Figures S1 and S2). Notably, in the study by Zhu et al.,¹⁶ maximum plasma concentrations of PFOS and PFOA at the high dose (5 mg/kg_{bw}) were unexpectedly higher following oral compared to IV administration,

despite identical doses. This difference was not observed at lower doses or sustained across the full concentration–time profile. These inconsistencies are likely driven by variability in the *in vivo* data rather than systematic model bias, as the PBTK model performed well at 1 mg/kg_{bw} IV doses with R^2 values of 0.77 and 0.83 for PFOS and PFOA, respectively.

Modeled elimination half-lives ($t_{1/2}$) for long-chain PFAA ($\eta_{\text{pfc}} \geq 7$) were consistent with experimental values. The modeled $t_{1/2}$ for PFOS was 42–58 days compared to an observed range of 31–72 days.^{10,16,19} For PFOA, the modeled $t_{1/2}$ range was 12–14 days compared to observed ranges of 12–42 days.¹⁶ For PFNA, the modeled range in $t_{1/2}$ was 27–55 days compared to an observed range of 26–69 days.¹⁵ The modeled $t_{1/2}$ for perfluorodecanoic acid (PFDA) was 48–66 days compared to an observed range between 52–66 days⁵⁸ (Figure 2 and Table 1). For PFHxS ($\eta_{\text{pfc}} = 6$), modeled $t_{1/2}$ was 28–34 days compared to experimental values of 25–54 days.^{16,17} In contrast, the model overestimated $t_{1/2}$ for short-chain PFAS ($\eta_{\text{pfc}} \leq 5$). PFBS ($\eta_{\text{pfc}} = 4$) and perfluorohexanoic acid (PFHxA, $\eta_{\text{pfc}} = 5$) had modeled $t_{1/2}$ values of 3.2 and 3.8 days, respectively, compared to experimental values of ~ 6 h. This discrepancy may reflect missing renal clearance mechanisms in the model. PFBS and PFHxA exhibit high affinity for OAT1 and OAT3,²³ indicating efficient uptake into renal proximal tubule cells and suggesting they may also be substrates for apical efflux transporters such as multidrug resistance-associated proteins, for which *in vitro* data are currently lacking.

Modeled PFOS concentrations in liver, kidneys, and the rest of the body fell within a factor of 3 of experimental observations across all dosing regimens. These results included single oral doses ranging from 0.1 to 5 mg/kg_{bw} (sampled after 65 days)¹⁹ and repeated daily oral doses of 0.5 or 10 mg/kg_{bw} over 7 days (Figure 2D).¹⁸ Some deviations in gut concentrations were observed, likely due to the assumption of constant oral uptake in the model, whereas mice were fed at defined intervals in the experimental studies. Tissue concentration data were only available for PFOS, limiting model evaluation for other PFAA.

The PBTK model successfully captured differences in PFOS distribution and elimination between knockout and wild-type mice. Previous studies showed that PFOS blood concentrations were not significantly different in FABP knockout and wild-type mice exposed to the same doses.¹⁹ Model results similarly indicated that deletion of liver-FABP and intestinal-FABP did not influence PFOS blood concentrations (Figure S5). While FABPs have been hypothesized to facilitate hepatic PFAS uptake based on *in vitro* studies,⁵³ our findings suggest their contribution to PFOS toxicokinetics *in vivo* is minimal. In knockout mice with global genetic deletion of albumin ($\text{Alb}^{-/-}$), observed plasma PFOS concentration were 52–65% lower and liver concentrations were 23–26% higher than wild-type mice.¹⁸ Modeled results predicted a similar 48% reduction in plasma PFOS concentrations, but lower increases compared to observed changes in liver concentrations (4%) (Figure S6). This discrepancy likely reflects the dominant role of phospholipid binding in hepatic PFOS retention,⁵⁹ which may reduce the impact of albumin loss. Overall, we find the model reasonably reproduced observed PFOS distribution patterns in knockout mice with global albumin and FABP deletion.

3.2. Simulating Absorption, Distribution, and Elimination for PFAA.

Model results illustrate a rapid increase in gut and liver concentrations of the nine PFAA examined in this study within the first 24 h following a single oral dose (Figures 3A and S7). Short-chain PFAA, such as PFBS ($\eta_{\text{pfc}} = 4$) and PFHxA ($\eta_{\text{pfc}} = 5$), reach peak concentrations in systemic blood, kidneys, and the rest compartment within 24 h, and decline thereafter (Figure S7A–C). In contrast, maximum blood and kidney concentrations for $\eta_{\text{pfc}} \geq 7$ PFAA exhibit a 2–3 day delay after dosing (Figure S7D–H). Accumulation of $\eta_{\text{pfc}} \geq 9$ PFAA in the rest compartment continued throughout the 15-day simulation, underscoring their prolonged tissue retention (Figure S7G,H).

To simulate constant exposure conditions relevant to humans, we modeled repeated daily dosing at 100 ng over two years. Under these conditions, short-chain carboxylates ($\eta_{\text{pfc}} = 4 - 6$), including PFBS, PFHxA, and PFHpA, accumulated to significantly lower body concentrations than long-chain sulfonates and carboxylates (Figure 3C). For example, PFDA ($\eta_{\text{pfc}} = 9$) accumulated to levels approximately 100 times higher than PFBS ($\eta_{\text{pfc}} = 4$). Tissue distribution simulations showed high relative concentrations of all PFAA in blood, liver, and kidneys (Figure 3C). Notably, PFHxS ($\eta_{\text{pfc}} = 6$) accumulated to higher tissue concentrations and masses than PFOA ($\eta_{\text{pfc}} = 7$), highlighting the greater bioaccumulation potential of sulfonates compared to carboxylates. Longer chain length PFAA ($\eta_{\text{pfc}} \geq 7$) exhibited higher concentrations in the liver compared to shorter chain length PFAA ($\eta_{\text{pfc}} \leq 6$), which were more concentrated in the kidneys. Despite low concentrations, the rest compartment was the dominant sink for PFAA mass due to its large volume (24 mL) (Figure S9).

The fraction of PFAA absorbed was calculated with respect to the total amount of each compound in all compartments relative to the administered dose. Absorption across all PFAA was high, ranging from 76% to 90% following a single oral dose (Figure 3D). Shorter chain length PFAA exhibited slightly lower absorption ($\eta_{\text{pfc}} \leq 5 = 76 - 83\%$) compared to longer chain length PFAA ($\eta_{\text{pfc}} \geq 6 = 84 - 90\%$). Between 70–92% of the total administered dose over two years was eliminated through urinary excretion across all PFAA, making it the primary elimination pathway. Fecal excretion was most prominent for $\eta_{\text{pfc}} = 5 - 7$ PFAA, peaking at 30% for PFHxS. Notably, PFOS exhibited the highest urinary excretion (92%) and the lowest fecal elimination (8%). Reduced fecal excretion of $\eta_{\text{pfc}} \geq 7$ PFAA such as PFOS (8%), PFDA (12%), and perfluoroundecanoic acid (PFUDA, 9%) is likely due to enterohepatic recirculation,²⁸ where compounds excreted into the gut via biliary transport are reabsorbed in the gut lumen because of their high permeability.

Intratissue distribution modeling revealed extensive binding of all PFAA to proteins and phospholipids, with short-chain PFAA such as PFBS and PFHxA bound at $\geq 98\%$ (Figures 3B and S8). In the liver, 7 of the 9 PFAA were predominantly bound to phospholipids, with PFOS and PFDA showing over 80% phospholipid binding, while the remaining fraction was mainly protein-bound (Figure 3B). A similar binding pattern was observed in the kidney, gut, and rest compartments, where phospholipid binding accounted for $\geq 50\%$ of

PFAA binding (Figure S8). PFAA with $\eta_{\text{pfc}} \geq 7$, such as PFOS and PFUnDA, demonstrated particularly strong phospholipid binding, exceeding 80–90%. In contrast, more than 50% of PFHxA ($\eta_{\text{pfc}} = 5$) and PFHpA ($\eta_{\text{pfc}} = 6$) were bound to proteins across all tissues. In blood, all PFAA except PFOS were bound predominantly to albumin and globulins, with binding exceeding 50% for all PFAA.

3.3. Sensitivity of Toxicokinetic Parameters.

PBTK model sensitivity analyses showed that elimination half-lives of $\eta_{\text{pfc}} \geq 8$ PFAA are strongly driven by phospholipid binding and permeability. For these compounds, normalized sensitivity coefficients – defined as the percent change in toxicokinetic outcome per percent change in model parameter – ranged from 47% to 83% for membrane lipids and –73% to –84% for permeability. In contrast, $\eta_{\text{pfc}} \leq 7$ PFAA elimination showed greater sensitivity to renal transporter activity, particularly Oat3, with coefficients of –37% for PFBS and –35% for PFHpA. The pronounced sensitivity to phospholipid binding for $\eta_{\text{pfc}} \geq 7$ PFAA is driven by their strong affinity for phospholipids ($\log K_{\text{PL/w}}: 3.51 - 4.88$)^{47,48} and the widespread presence of phospholipids in mouse tissues such as the liver (2.05%) and kidneys (2.06%).^{21,32} This agrees with intratissue simulations showing that most PFAA are primarily bound to phospholipids in tissues (Figure 3B).

Permeability acts as a rate-limiting factor for PFAS distribution from central blood to the kidneys and liver, which facilitates renal and biliary excretion. PFAA with $\eta_{\text{pfc}} \geq 7$ exhibit higher apparent permeabilities, with values of 1.40×10^{-5} cm/s for PFOS and 8.44×10^{-6} cm/s for PFDA,^{23,47} reducing their reliance on renal transporters for excretion. In contrast, $\eta_{\text{pfc}} < 7$ PFAA have lower permeabilities, with values of 1.38×10^{-6} cm/s for PFBS and 1.16×10^{-6} cm/s for PFHxA, increasing their dependence on renal transporters.²³

Albumin is the strongest driver of PFAA accumulation in blood of the tested model parameters with sensitivity coefficients ranging from 21% to 64% (Figure 4B), due to its dual role as the most abundant protein in mouse blood (1.43% by volume)³² and its strong binding affinity for PFAA ($\log K_{\text{SA/w}}: 3.61 - 4.71$).^{46,49,52,53} In contrast, albumin is minimally present in tissues, with concentrations as low as 0.12% in the liver and 0.14% in the kidneys, therefore retaining PFAA in the blood and reducing distribution to peripheral tissues. Globulins, the second most abundant class of blood proteins (1.33% by volume)³² have a minor yet compound-specific impact on blood AUC. For $\eta_{\text{pfc}} \leq 9$ PFAA, sensitivity coefficients for globulin binding remain below 7%, but globulins contribute more significantly to blood levels of PFUDA ($\eta_{\text{pfc}} = 10$, sensitivity: 26%), consistent with their reported affinity to bind $\eta_{\text{pfc}} \geq 10$ PFAS.⁴⁶ Phospholipids negatively influence blood AUC by promoting PFAA retention in tissues, with sensitivity coefficients of –25% for PFOS. For $\eta_{\text{pfc}} \leq 6$ PFAA, permeability positively affects blood AUC due to the contribution of absorption within the simulated one elimination half-life. Conversely, for long-chain PFAA, permeability negatively affects blood AUC as it promotes renal elimination.

Structural proteins, such as actin, myosin, and collagen, are highly abundant in tissues, constituting up to 21% of the liver and 18% of the kidney in mice (Table S6).³² However, their contribution to PFAA accumulation in tissues is limited, with sensitivity coefficients ranging from 1.2%–16%, compared to 8%–32% for phospholipids (Figure S10). This difference is likely explained by the stronger binding affinity of PFAA to phospholipids^{47,48} compared to structural proteins ($\log K_{SP/W}$: 0.64 – 3.42).³² FABPs, previously suggested as key contributors to hepatic PFAA accumulation,⁵³ show minimal sensitivity across absorption, distribution, and elimination end points, with coefficients consistently below 3%. Their limited impact is explained by their low abundance in tissues, with 0.25% in liver and 0.31% in kidney by volume,³² and their relatively low binding strength for PFAA ($\log K_{FABP/W}$: 2.30 – 3.94).^{53–55}

Oatps and Ntcp only minorly influence PFAA toxicokinetics across absorption, blood and tissue retention, and elimination, with sensitivity coefficients reaching up to 6% for absorption (Figure S11). These transporters are located in the liver, where they facilitate the active uptake of PFAA from the hepatic portal vein and hepatic artery into hepatocytes.^{24,26,50} In addition, Oatp2b1 is present on the apical membrane of enterocytes, facilitating PFAA influx from the gut lumen. These transport processes promote PFAA distribution into the systemic circulation, retention in the liver, and biliary transport, contributing to enterohepatic recirculation.²⁸ Consequently, an increase in Oatp or Ntcp-mediated transport enhances absorption, tissue retention (particularly in the liver), and elimination half-lives, while decreasing blood AUC.

Oat-mediated transport influenced PFAA blood levels and clearance, particularly for $\eta_{pfc} \leq 7$ PFAA, with sensitivity coefficients for Oat1 and Oat3 ranging from –7.7% (Oat1) to 44% (Oat3) (Figure 4). By contrast, Oat2 had minimal impact on toxicokinetic outcomes for all PFAA, with sensitivity coefficients consistently below 1%. Located in the kidneys, Oat1–3 transport PFAA from the blood into renal tissues, thereby promoting renal elimination. For $\eta_{pfc} \geq 8$ PFAA, sensitivity coefficients for Oats ranged from –1.4% to –5%, indicating negligible to moderate importance of these transporters and highlighting the role of passive permeability for these highly permeable PFAA.²³

3.4. Implications, Limitations, and Applicability to Humans.

Several PBPK models have been developed to describe PFAS kinetics in rodents and humans.^{60–62} These models rely on empirically fitted partition coefficients and clearance terms to simulate tissue distribution and elimination. In contrast, our model uses *in vitro*-derived parameters, without calibration to *in vivo* data. This structure allows for mechanistic evaluation of individual toxicokinetic mechanisms and facilitates refinement where key pathways – such as transporter-mediated renal clearance of short-chain PFAS – remain poorly defined.

Our PBTK model accurately predicts toxicokinetic end points for most PFAA in mice across single and repeated oral and intravenous dosing regimens, with R^2 values ≥ 0.65 and rRMSE $\leq 122\%$. Exceptions included underestimation of early PFHxS plasma concentrations after oral dosing and overestimation of PFOS and PFOA plasma levels at high IV doses of 5

mg/kg_{bw}.¹⁶ This was potentially due to saturation of clearance pathways, as oral dosing at the same concentration was well predicted. The model reproduces observed changes in PFAA exposures in FABP- and albumin-deficient knockout mice. Additional tissue concentration data are needed to assess the performance of the PBTK model for PFAA other than PFOS. The model overestimated elimination half-lives of short-chain PFBS ($\eta_{pfc} = 4$) and PFHxA ($\eta_{pfc} = 5$), highlighting the need for refining renal clearance mechanisms in the model to better capture their rapid elimination. Emerging rodent and human toxicokinetic data on short-chain PFAS alternatives such as GenX⁶³ and F-53B⁶⁴ can be combined with *in vitro*-derived permeability, binding, and transporter parameters in our PBTK model to identify compound-specific mechanisms driving elimination half-lives and tissue distribution of emerging PFAS.

Sensitivity analyses revealed that permeability and phospholipid binding were the most important model parameters for $\eta_{pfc} \geq 7$ PFAA, while renal transporters, particularly Oat3, substantially impacted the clearance of $\eta_{pfc} < 7$ PFAA. Albumin and phospholipids were the primary determinants of PFAA retention in blood and tissues, respectively. Structural proteins played a minor role in tissue retention, while FABPs showed minimal sensitivity to tissue accumulation and elimination. Interestingly, PFAA absorption was the toxicokinetic outcome that was least influenced by shifts in model parameters (Figure S11). Alongside Oats and Oatps, transporters like the fatty acid translocase or platelet glycoprotein 4 (CD36) and the breast cancer resistance protein (Bcrp) may contribute to PFAS toxicokinetics,^{65,66} but their roles in absorption, distribution, and elimination remain unclear due to limited quantitative data. Transporter data used in the current model version were from *in vitro* cells genetically modified to overexpress the transporter of interest and literature scaling factors.⁵⁷ A source of uncertainty in the current model is the absence of experimentally determined Michaelis–Menten parameters for the included membrane transporters, including K_m (affinity) and V_{max} (maximum transport rate), which limits incorporation of transporter saturation kinetics into uptake and efflux processes. In addition, limited data on transporter expression in the *in vitro* cells used to generate these kinetics constrains accurate *in vitro* to *in vivo* scaling. Addressing these gaps through targeted experiments and quantitative proteomics could enhance model accuracy, especially for short-chain PFAS with transporter-dependent kinetics.

Expanding the applicability domain of the model requires inclusion of additional exposure routes, including inhalation and dermal absorption. The ongoing shift from legacy to new PFAS with fewer fluorinated carbons and distinct functional groups emphasizes the need for kinetic and binding data on emerging PFAS measured *in vitro*. The current model was developed using uniform physiological parameters for laboratory mice. Future applications in humans should incorporate interindividual variability in transporter and protein expression, as well as physiological changes associated with genotype, age, or disease states. As shown in prior work, variability in blood protein concentrations alone can cause up to 3-fold differences in free PFAS fractions, potentially affecting blood elimination and tissue accumulation.⁴⁶ Noninvasive techniques, such as liquid biopsies,⁶⁷ could provide personalized physiological data, enabling the development of individualized human PBTK models. Such models could be used to explore how disease and genetic

factors alter PFAS toxicokinetics. For example, fatty liver disease is associated with elevated PFAS exposures,⁶⁸ but it remains uncertain whether PFAS triggers fatty liver disease or if the condition itself increases PFAS accumulation in the liver. Reduced blood albumin levels observed in fatty liver disease⁶⁹ could decrease PFAS retention in blood and increase hepatic uptake. This mechanism is well-documented for highly albumin-bound drugs like warfarin, where reduced albumin increases the unbound fraction, elevating anticoagulant activity and risk for bleeding.⁷⁰ Incorporating variations in PFAS exposures, physiological changes (e.g., reduced albumin), and toxicological targets into a PBTK/TD (toxicodynamic) framework can help disentangle these interactions and improve understanding of the relationships between exposure and health outcomes. Future applications of this approach could aid in identifying risk factors associated with elevated chemical accumulation and heightened susceptibility to toxicity, while supporting the development of targeted intervention strategies.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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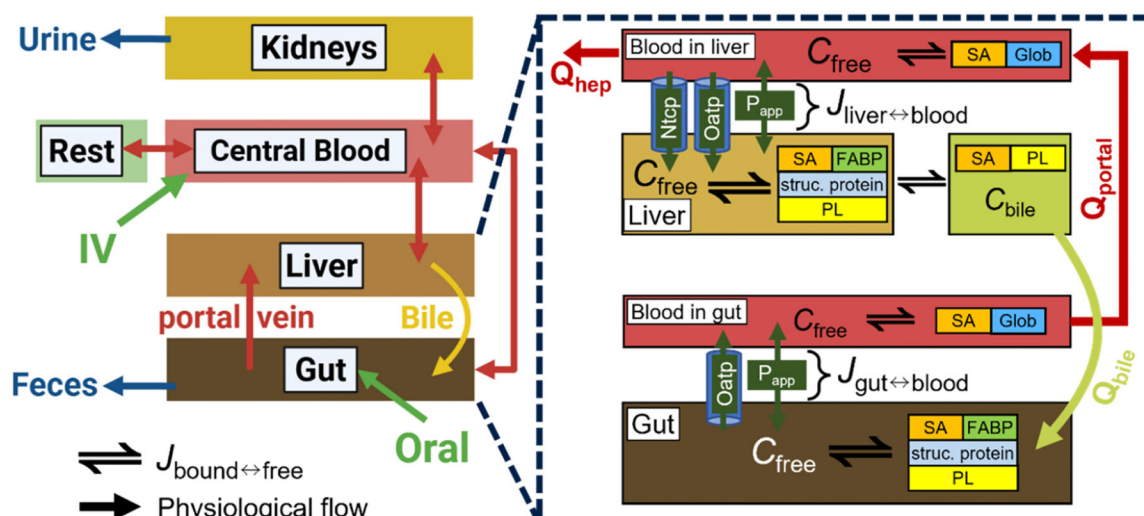
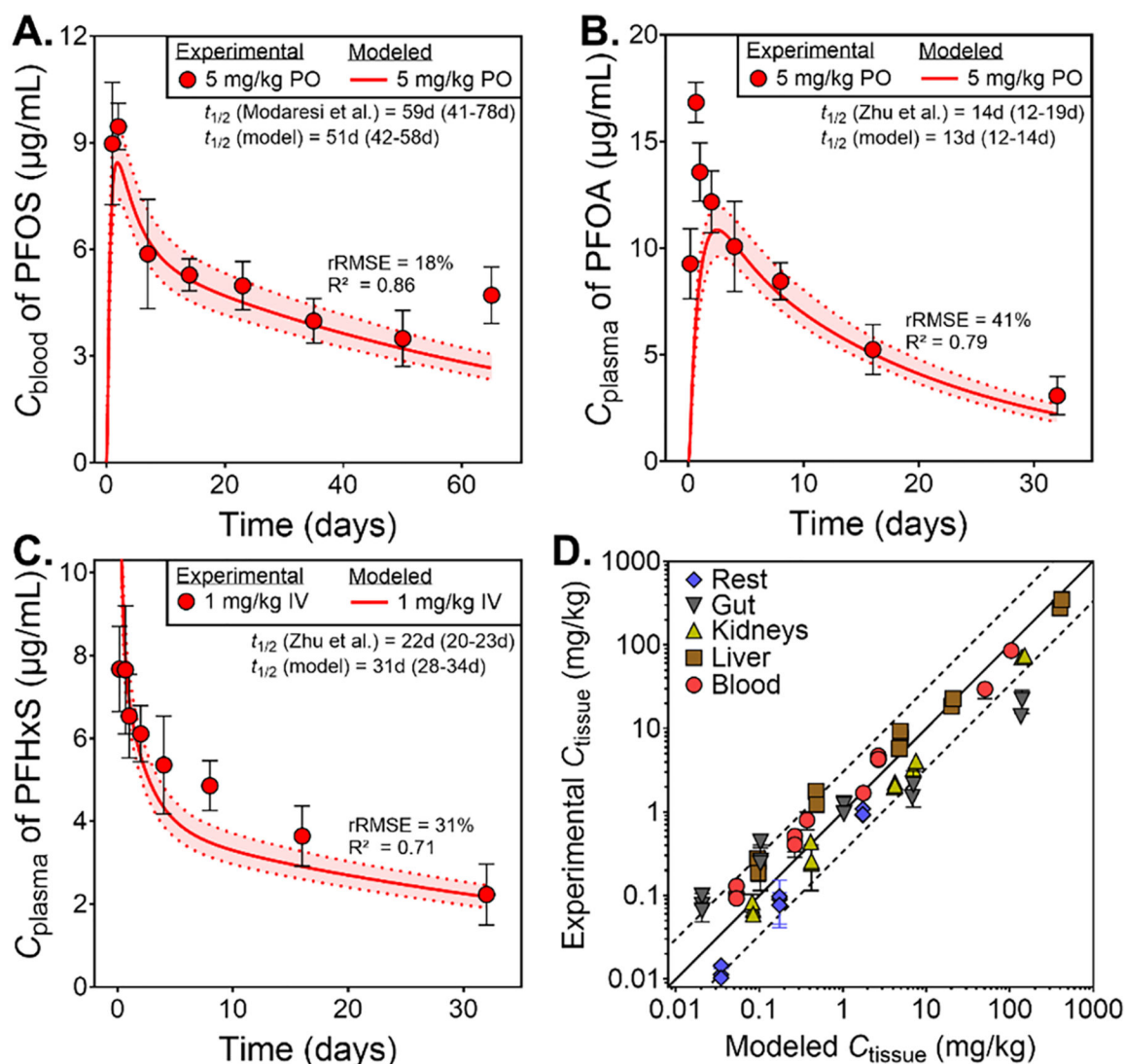
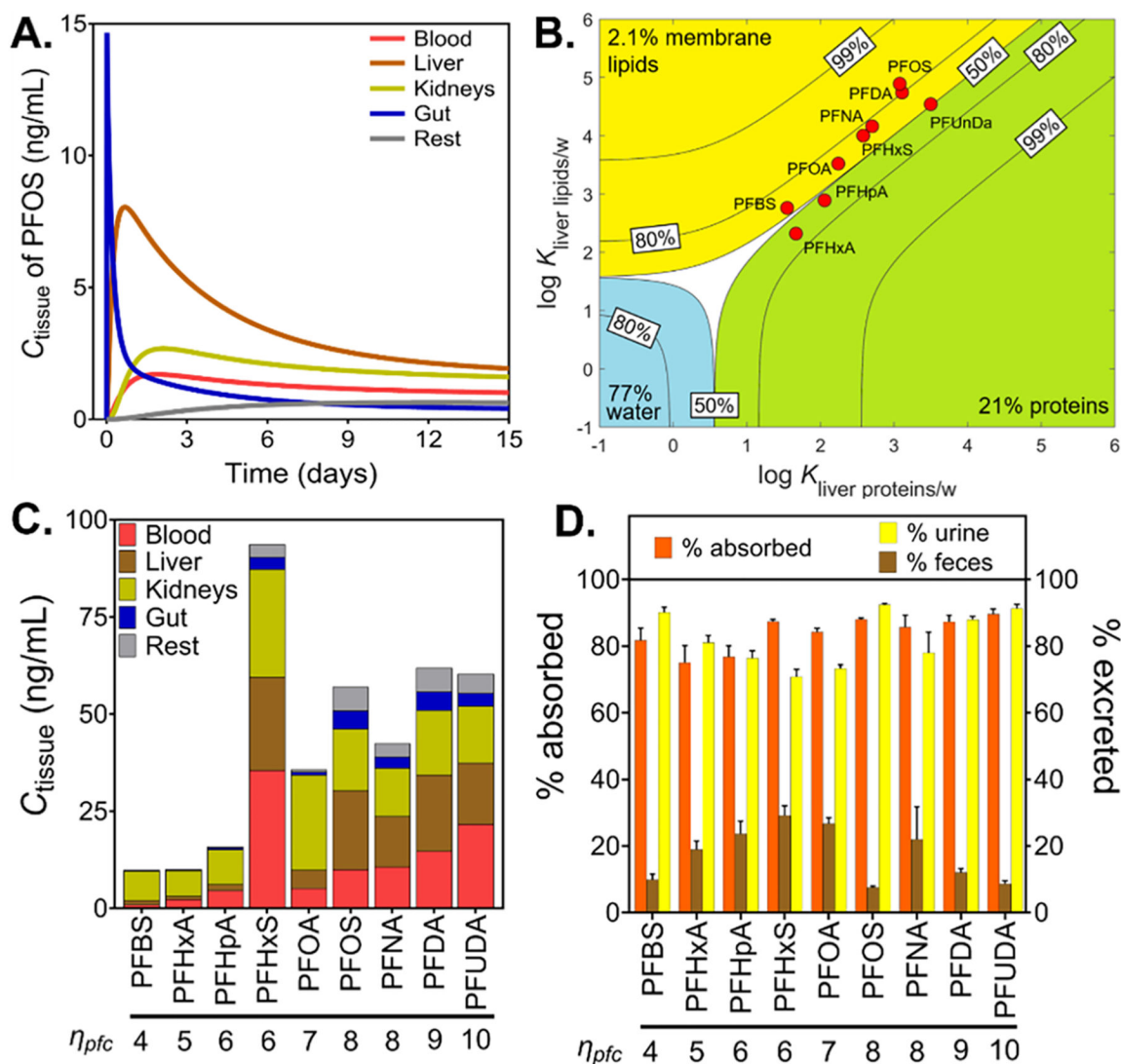


Figure 1.

Schematic for the five-compartment PBTK model developed in this study that includes blood, gut, liver, kidney, and rest of body. Intratissue distribution is driven by binding to proteins and phospholipids. Blood transport, intertissue distribution, and excretion are based on physiological flow rates (Q) and chemical transport rates (permeability, membrane transporters). Rest: nonspecified organs; Q_{hep} : hepatic artery blood flow; Q_{portal} : portal vein blood flow; Q_{bile} : bile flow rate; C_{free} : unbound chemical concentration in blood or tissue; C_{bile} : chemical concentration in bile; P_{app} : apparent membrane permeability; $Ntcp$: sodium taurocholate cotransporting polypeptide; $Oatp$: organic anion-transporting polypeptide; SA : serum albumin; $Glob$: globulins; $FABP$: fatty acid-binding protein; $Struc. protein$: structural proteins; PL : membrane phospholipids.

**Figure 2.**

Evaluation of the physiologically based toxicokinetic model (PBTK) model developed in this study against observations from three experimental mouse studies. Panels (A–C) compare modeled and experimental blood or plasma concentrations of PFOS (A), PFOA (B), and PFHxS (C) over time in C57BL/6J (A) or CD-1 mice (B, C) following single oral (PO) or intravenous (IV) doses. Modeled and experimental elimination half-lives with 95% confidence intervals, along with $r\text{RMSE}$ and R^2 are provided in each plot. Panel (D) compares modeled and experimental PFOS concentrations across blood, liver, kidney, gut, and rest tissues after single or multiple oral doses, with solid and dashed lines representing 1:1, 3:1, and 1:3 agreement.

**Figure 3.**

Panel (A) shows modeled temporal concentrations of PFOS across blood, liver, kidney, gut, and rest tissues following a single oral dose. Panel (B) illustrates the distribution of 9 PFAA among proteins, phospholipids, and water in mouse liver, with lines indicating 50%, 80%, and 99% distribution thresholds. Panel (C) compares concentrations of 9 PFAA in blood, liver, kidney, gut, and rest tissue following a repeated daily dose over 2 years. Panel (D) shows the percentage of PFAA absorbed, excreted in urine, and excreted in feces following a single oral exposure and after two years of simulation.

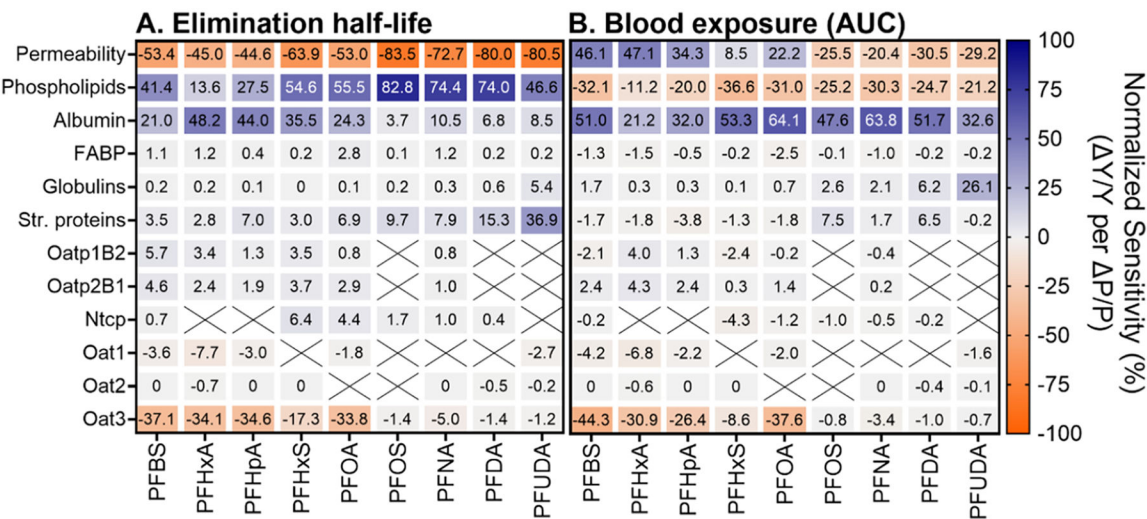


Figure 4. Sensitivity analysis of toxicokinetic parameters for (A) elimination half-life and (B) blood exposure (AUC) of nine PFAA in mice. Parameters were perturbed by +10%, and normalized sensitivity coefficients were calculated as the ratio of the relative change in the outcome *Y* to the relative change in the parameter *P* (blue: positive, orange: negative). Elimination half-life analyses used single oral doses with durations of ten half-lives of each PFAA, while AUC analyses spanned one half-life to capture absorption, distribution, and elimination. Diagonal lines indicate toxicokinetic parameters not included for a given PFAA due to lack of measurable transporter-mediated uptake *in vitro*.

Table 1.
Modeled (This Study) and Experimental Elimination Half-Lives ($t_{1/2}$)

chemical ^a	modeled $t_{1/2}$ (95% CI) ^b	measured $t_{1/2}$ (range in literature) ^b
PFBS (C4-SA)	3.2d (2.7–4.0d)	5.8h ¹³
PFHxA (C5-CA)	3.8d (3.2–5.2d)	<6h ¹⁴
PFHpA (C6-CA)	6.9d (5.9–8.1d)	N/A
PFHxS (C6-SA)	31d (28–34d)	25–54d ^{16,17}
PFOA (C7-CA)	13d (12–14d)	12–42d ¹⁶
PFOS (C8-SA)	51d (42–58d)	31–72d ^{10,16,19}
PFNA (C8-CA)	38d (27–55d)	26–69d ¹⁵
PFDA (C9-CA)	56d (48–66d)	52–66d (rat) ⁵⁸
PFUDA (C10-CA)	47d (28–88d)	N/A

^aPerfluoroalkyl acids (carboxylic acids: CA or sulfonic acids: SA) are indicated by their carbon chain lengths in parentheses (e.g., C4 = four perfluorinated carbons).

^bRanges indicate 95% confidence intervals (95% CI) determined by probabilistic modeling to experimentally measured $t_{1/2}$ ranges from various studies for seven PFAA (N/A = not available).